

# Horse Race

## Green Book 2.2: Logic Reasoning

**Problem.** There are 25 horses, each of which runs at a constant speed that is different from the other horses' speeds. Since the track only has 5 lanes, each race can have at most 5 horses. If we need to find the 3 fastest horses, what is the minimum number of races needed to identify them?

**Solution.**

First divide the 25 horses into 5 groups of 5:

(1, 2, 3, 4, 5), (6, 7, 8, 9, 10), (11, 12, 13, 14, 15), (16, 17, 18, 19, 20), (21, 22, 23, 24, 25).

Race each group. After these 5 races, we know the order within each group. WLOG, assume the order in each group follows the numbering:

$$1 > 2 > 3 > 4 > 5,$$

$$6 > 7 > 8 > 9 > 10,$$

$$11 > 12 > 13 > 14 > 15,$$

$$16 > 17 > 18 > 19 > 20,$$

$$21 > 22 > 23 > 24 > 25.$$

Here  $x > y$  means horse  $x$  is faster than horse  $y$ .

The bottom two horses in each group cannot be in the top 3 overall, because each of them already has at least 3 horses in its own group faster than it. Thus we may immediately eliminate

$$4, 5, \quad 9, 10, \quad 14, 15, \quad 19, 20, \quad 24, 25.$$

Now race the 5 winners:

$$1, 6, 11, 16, 21.$$

WLOG, suppose the result is

$$1 > 6 > 11 > 16 > 21.$$

Then horse 1 is the fastest overall.

Also, any horse in the groups headed by 16 or 21 cannot be in the top 3, because even the fastest horse in those groups is slower than 1, 6, and 11. So we eliminate

$$16, 17, 18, \quad 21, 22, 23.$$

Now consider the group headed by 11. Since

$$1 > 6 > 11,$$

horse 11 can still be third overall, but no horse behind 11 in its group can be top 3. So we eliminate

12, 13.

Next consider the group headed by 6. Since

$1 > 6$ ,

horse 6 can be second overall, and horse 7 can still be third overall. But horses behind 7 cannot be top 3, so we eliminate

8.

Thus the only possible candidates for second and third place are

2, 3, 6, 7, 11.

Horse 1 is already known to be the fastest, so one final race among

2, 3, 6, 7, 11

determines the second and third fastest horses.

Therefore the minimum number of races needed is

$$5 + 1 + 1 = 7.$$

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